

Physics 523 HW 1

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1. 1

I use the notation, technology, and results I developed in my Real Scalar Field Notes, which are largely based on Schwartz.

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 \quad (1)$$

$$2\mathcal{L}_0 = -\Phi(\square + m^2)\Phi, \quad \mathcal{L}_1 = -\frac{\lambda}{4!}\Phi^4 \quad (2)$$

$$S = \int d^4x \mathcal{L} = S_0 + S_1 \quad (3)$$

$$S_0 = \int d^4x \mathcal{L}_0, \quad S_1 = \int d^4x \mathcal{L}_1 \quad (4)$$

The generating functional is

$$Z[J] = \int \mathcal{D}\Phi(\vec{y}) e^{i(S+I)}, \quad I[J] = \int d^4x J(\vec{x})\Phi(\vec{x}). \quad (5)$$

The free field generating functional is

$$Z_0[J] = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)}. \quad (6)$$

In closed form

$$Z_0[J] = e^{-iH[J]/2}, \quad H[J] = \int d^4x d^4y J(\vec{x})\mathcal{G}(\vec{x} - \vec{y})J(\vec{y}), \quad (7)$$

where the unbounded (it lacks a boundary condition) Green's function is

$$(2\pi)^4 \mathcal{G}(\vec{x} - \vec{y}) = - \int d^4p \frac{e^{i\vec{p} \cdot (\vec{x} - \vec{y})}}{p^2 - m^2}. \quad (8)$$

I have dropped the formally infinite constant since it will not appear in the final result.

$$Z[J] = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+S_1+I)} = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} e^{iS_1} \quad (9)$$

I expand in small λ

$$e^{iS_1} = 1 + iS_1 + \mathcal{O}(\lambda^2). \quad (10)$$

$$e^{iS_1} = 1 + i \int d^4x \mathcal{L}_1 = 1 - \frac{i\lambda}{4!} \int d^4x \Phi(\vec{x})^4 \quad (11)$$

$$e^{iS_1} = 1 - \frac{i\lambda}{24} \int d^4x \Phi(\vec{x})^4 \quad (12)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \left[1 - \frac{i\lambda}{24} \int d^4x \Phi(\vec{x})^4 \right] \quad (13)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} - \frac{i\lambda}{24} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x \Phi(\vec{x})^4 \quad (14)$$

$$Z = Z_0 - \frac{i\lambda}{24} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x \Phi(\vec{x})^4 \quad (15)$$

$$Z = Z_0 - \frac{i\lambda}{24} \int d^4x \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x})^4 \quad (16)$$

$$-i \frac{\delta Z_0}{\delta J(\vec{x})} = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x}) \quad (17)$$

$$(-i)^n \frac{\delta^n Z_0}{\delta J(\vec{x})^n} = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x})^n \quad (18)$$

$$Z = Z_0 - \frac{i\lambda}{24} \int d^4x \frac{\delta^4 Z_0}{\delta J(\vec{x})^4} \quad (19)$$

$$Z[J] = Z_0[J] - \frac{i\lambda}{24} \int d^4x \frac{\delta^4 Z_0[J]}{\delta J(\vec{x})^4} \quad (20)$$

$$\langle \Omega | T \Phi(\vec{x}_1) \Phi(\vec{x}_2) | \Omega \rangle = - \frac{1}{Z[0]} \frac{\delta^2 Z[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \Big|_{J=0} \quad (21)$$

$$Z[0] = Z_0[0] - \frac{i\lambda}{24} \int d^4x \frac{\delta^4 Z_0[J]}{\delta J(\vec{x})^4} \Big|_{J=0} \quad (22)$$

$$H[0] = 0 \implies Z_0[0] = e^{-iH[0]/2} = e^0 = 1 \quad (23)$$

$$Z[0] = 1 - \frac{i\lambda}{24} \int d^4x \frac{\delta^4 Z_0[J]}{\delta J(\vec{x})^4} \Big|_{J=0} \quad (24)$$

I use the shorthands

$$\delta_x = \frac{\delta}{\delta J(\vec{x})}, \quad \delta_{xx} = \delta_{x^2} = \frac{\delta^2}{\delta J(\vec{x})^2} \quad (25)$$

$$Z[0] = 1 - \frac{i\lambda}{24} \int d^4x (\delta_{x^4} Z_0[J])_{J=0} \quad (26)$$

$$(\delta_{x^4} Z_0[J])_{J=0} = -\frac{3}{4} Z_0[0] (\delta_{xx} H[J])_{J=0}^2 = -\frac{3}{4} [\delta_{xx} H]_{J=0}^2 \quad (27)$$

There are more derivatives of H , but as I show in my Scalar Field Notes, only terms composed entirely of second derivatives survive. This is the path integral analogue of only fully contracted terms surviving in canonical quantization. In my Scalar Field Notes I found

$$\delta_{xy} H = 2\mathcal{G}^{xy} = 2\mathcal{G}(\vec{x} - \vec{y}) \quad (28)$$

$$(\delta_{x^4} Z_0[J])_{J=0} = -3(\mathcal{G}^{xx})^2 \quad (29)$$

$$Z[0] = 1 + \frac{i}{8} \lambda \int d^4x (\mathcal{G}^{xx})^2 \quad (30)$$

Notice that $\mathcal{G}^{xx} = \mathcal{G}(\vec{x} - \vec{x}) = \mathcal{G}(0)$. But writing $\mathcal{G}^{xx}$ as I do here is standard practice. The last step as I explain in my Scalar Field Notes is to replace the unbounded propagator with Feynman propagator using the rule

$$\mathcal{G} \rightarrow -i\mathcal{G}_F, \quad (31)$$

where

$$(2\pi)^4 \mathcal{G}_F(\vec{x} - \vec{y}) = i \lim_{\epsilon \rightarrow 0^+} \int d^4p \frac{e^{i\vec{p} \cdot (\vec{y} - \vec{x})}}{\vec{p}^2 - m^2 + i\epsilon}. \quad (32)$$

$$Z[0] = 1 - \frac{i}{8} \lambda \int d^4x (\mathcal{G}_F^{xx})^2 \quad (33)$$

I define

$$\mathfrak{F} = \int d^4x (\mathcal{G}_F^{xx})^2, \quad (34)$$

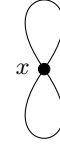


FIG. 1. Feynman diagram for $\mathfrak{F}$.

which corresponds to the bubble diagram in Fig. 1.

$$Z[0] = 1 - \frac{i}{8} \lambda \mathfrak{F} \quad (35)$$

$$\langle \Omega | T \Phi(\vec{x}_1) \Phi(\vec{x}_2) | \Omega \rangle = - \frac{1}{Z[0]} \frac{\delta^2 Z[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \Big|_{J=0} \quad (36)$$

I use the shorthands

$$\Phi(\vec{x}_1) = \Phi_1, \quad \delta_1 = \frac{\delta}{\delta J(\vec{x}_1)} \quad (37)$$

$$\langle \Omega | T \Phi_1 \Phi_2 | \Omega \rangle = - \frac{(\delta_{12} Z[J])_{J=0}}{Z[0]} \quad (38)$$

I use another shorthand

$$(\delta_{12} Z[J])_{J=0} = Z_{12}[0], \quad (39)$$

because it looks nicer.

$$\langle \Omega | T \Phi_1 \Phi_2 | \Omega \rangle = - \frac{Z_{12}[0]}{Z[0]} \quad (40)$$

$$Z_{12}[J] = \delta_{12} Z_0[J] - \frac{i\lambda}{24} \int d^4x \delta_{12} \frac{\delta^4 Z_0[J]}{\delta J(\vec{x})^4} \quad (41)$$

$$Z_{12}[J] = \delta_{12} Z_0[J] - \frac{i\lambda}{24} \int d^4x \delta_{12x^4} Z_0[J] \quad (42)$$

$$Z_{12}[0] = (\delta_{12} Z_0[J])_{J=0} - \frac{i\lambda}{24} \int d^4x (\delta_{12x^4} Z_0[J])_{J=0} \quad (43)$$

$$\delta_{12} Z_0[J] = \delta_{12} e^{-iH[J]/2} \quad (44)$$

$$\delta_{12} Z_0[J] = - \frac{Z_0[J]}{2} \left[\frac{1}{2} \frac{\delta H[J]}{\delta J(\vec{x}_1)} \frac{\delta H[J]}{\delta J(\vec{x}_2)} + i \frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \right] \quad (45)$$

$$(\delta_{12} Z_0[J])_{J=0} = - \frac{Z_0[0]}{2} \left[\frac{1}{2} \frac{\delta H[J]}{\delta J(\vec{x}_1)} \frac{\delta H[J]}{\delta J(\vec{x}_2)} + i \frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \right]_{J=0} \quad (46)$$


 FIG. 2. $\mathfrak{V}[0]$

$$Z_0[0] = 1 \quad (47)$$

$$(\delta_{12}Z_0[J])_{J=0} = -\frac{1}{2} \left[\frac{1}{2} \frac{\delta H[J]}{\delta J(\vec{x}_1)} \frac{\delta H[J]}{\delta J(\vec{x}_2)} + i \frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \right]_{J=0} \quad (48)$$

From my Real Scalar Field Notes

$$\frac{\delta H[J]}{\delta J(\vec{x}_1)} = 2 \int d^4x \mathcal{G}(\vec{x} - \vec{x}_1) J(\vec{x}) \implies \frac{\delta H[J]}{\delta J(\vec{x}_1)} \Big|_{J=0} = 0. \quad (49)$$

$$(\delta_{12}Z_0[J])_{J=0} = -\frac{1}{2} \left[\frac{1}{2} \frac{\delta H[J]}{\delta J(\vec{x}_1)} \frac{\delta H[J]}{\delta J(\vec{x}_2)} + i \frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \right]_{J=0} \quad (50)$$

$$(\delta_{12}Z_0[J])_{J=0} = -\frac{i}{2} \frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \quad (51)$$

$$\frac{\delta^2 H[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} = \delta_{12}H = 2\mathcal{G}^{12} \quad (52)$$

$$\mathcal{G}^{12} = \mathcal{G}(\vec{x}_1 - \vec{x}_2) \quad (53)$$

$$(\delta_{12}Z_0[J])_{J=0} = -i\mathcal{G}^{12} \quad (54)$$

I replace the unbounded propagator with the Feynman propagator (Eq. 31).

$$(\delta_{12}Z_0[J])_{J=0} = -\mathcal{G}_F^{12} \quad (55)$$

I define (Fig. 2)

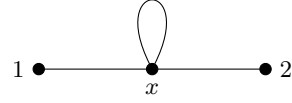
$$\boxed{\mathfrak{V}[0] = \mathcal{G}_F^{12}}. \quad (56)$$

$$(\delta_{12}Z_0[J])_{J=0} = -\mathfrak{V}[0] \quad (57)$$

$$Z_{12}[0] = -\mathfrak{V}[0] - \frac{i\lambda}{24} \int d^4x (\delta_{12x^4}Z_0[J])_{J=0} \quad (58)$$

Only second derivatives survive so

$$(\delta_{12x^4}Z_0[J])_{J=0} = \frac{3}{2}iZ_0[0](\delta_{xx}H)_{J=0} \left[(\delta_{x2}H)(\delta_{x1}H) + \frac{1}{4}(\delta_{xx}H)(\delta_{12}H) \right]_{J=0}. \quad (59)$$


 FIG. 3. $\mathfrak{V}[1]$

$$(\delta_{12x^4}Z_0[J])_{J=0} = \frac{3}{2}i(\delta_{xx}H)_{J=0} \left[(\delta_{x2}H)(\delta_{x1}H) + \frac{1}{4}(\delta_{xx}H)(\delta_{12}H) \right]_{J=0} \quad (60)$$

I give the rule for taking derivatives of H in Eq. 28.

$$(\delta_{12x^4}Z_0[J])_{J=0} = \frac{3}{2}i(\delta_{xx}H) \left[(\delta_{x2}H)(\delta_{x1}H) + \frac{1}{4}(\delta_{xx}H)(\delta_{12}H) \right] \quad (61)$$

$$(\delta_{12x^4}Z_0[J])_{J=0} = 3i\mathcal{G}^{xx}(4\mathcal{G}^{x2}\mathcal{G}^{x1} + \mathcal{G}^{xx}\mathcal{G}^{12}) \quad (62)$$

$$(\delta_{12x^4}Z_0[J])_{J=0} = -3\mathcal{G}_F^{xx}(4\mathcal{G}_F^{x2}\mathcal{G}_F^{x1} + \mathcal{G}_F^{xx}\mathcal{G}_F^{12}) \quad (63)$$

$$Z_{12}[0] = -\mathfrak{V}[0] + \frac{i}{8}\lambda \int d^4x \mathcal{G}_F^{xx}(4\mathcal{G}_F^{x2}\mathcal{G}_F^{x1} + \mathcal{G}_F^{xx}\mathcal{G}_F^{12}) \quad (64)$$

$$Z_{12}[0] = -\mathfrak{V}[0] + \frac{i}{2}\lambda \int d^4x \mathcal{G}_F^{x1}\mathcal{G}_F^{x2}\mathcal{G}_F^{xx} + \frac{i}{8}\lambda \mathcal{G}_F^{12} \int d^4x (\mathcal{G}_F^{xx})^2 \quad (65)$$

$$Z_{12}[0] = -\mathfrak{V}[0] + \frac{i}{2}\lambda \int d^4x \mathcal{G}_F^{x1}\mathcal{G}_F^{x2}\mathcal{G}_F^{xx} + \frac{i}{8}\lambda \mathfrak{V}[0]\mathfrak{V} \quad (66)$$

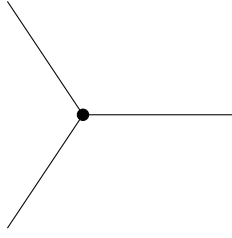
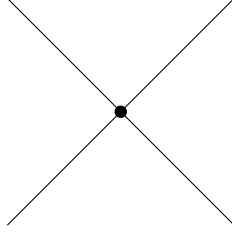
$$\boxed{\mathfrak{V}[1] = \int d^4x \mathcal{G}_F^{x1}\mathcal{G}_F^{x2}\mathcal{G}_F^{xx}} \quad (67)$$

I show the diagram for $\mathfrak{V}[1]$ in Fig. 3.

$$\boxed{Z_{12}[0] = -\mathfrak{V}[0] + \frac{i}{2}\lambda \left(\mathfrak{V}[1] + \frac{\mathfrak{V}}{4}\mathfrak{V}[0] \right)} \quad (68)$$

I plug Eqs. 35 & 68 into Eq. 40

$$\langle \Omega | T \Phi_1 \Phi_2 | \Omega \rangle = \left(1 - \frac{i}{8}\lambda \mathfrak{V} \right)^{-1} \left[\mathfrak{V}[0] - \frac{i}{2}\lambda \left(\mathfrak{V}[1] + \frac{\mathfrak{V}}{4}\mathfrak{V}[0] \right) \right]. \quad (69)$$


 FIG. 4. The 3-interaction vertex factor is $-ig$.

 FIG. 5. The 4-interaction vertex factor is $-i\lambda$.

I expand to first order in λ

$$\langle \Omega | T \Phi_1 \Phi_2 | \Omega \rangle = \mathfrak{G}[0] - \frac{i}{2} \lambda \mathfrak{G}[1]. \quad (70)$$

The bubble diagram $\mathfrak{Y}$ canceled out as expected.

2. 2

I use the notation I developed in my Real Scalar Field Notes. The Lagrangian is

$$\mathcal{L} = -\frac{\Phi(\Box + m^2)\Phi}{2} - \frac{\lambda}{4!}\Phi^4 - \frac{g}{3!}\Phi^3. \quad (71)$$

2.1. Feynman Rules

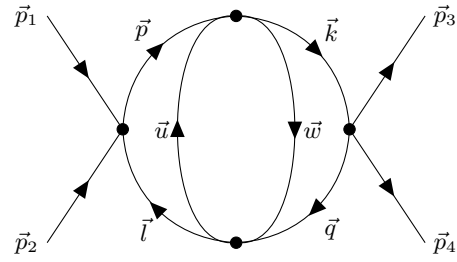
The propagator for virtual particles is

$$\frac{i}{\vec{p}^2 - m^2 + i\epsilon}. \quad (72)$$

I show the interaction vertices in Figs. 4 & 5.

2.2. a

I label this diagram $\star$ and show it in Fig. 6. Using my Real Scalar Field Notes I translate this diagram into an


 FIG. 6. $\star$

integral,

$$\begin{aligned} \star &= \frac{(-i\lambda)^4}{(2\pi)^{4n}} S \int d^4p d^4k d^4q d^4l d^4w d^4u \frac{i}{\vec{p}^2 - m^2 + i\epsilon} \\ &\quad \frac{i}{\vec{k}^2 - m^2 + i\epsilon} \frac{i}{\vec{q}^2 - m^2 + i\epsilon} \frac{i}{\vec{l}^2 - m^2 + i\epsilon} \frac{i}{\vec{w}^2 - m^2 + i\epsilon} \\ &\quad \frac{i}{\vec{u}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 + \vec{p}_2 + \vec{l} - \vec{p}) \delta^{(4)}(\vec{p} + \vec{u} - \vec{k} - \vec{w}) \\ &\quad \delta^{(4)}(\vec{k} - \vec{q} - \vec{p}_3 - \vec{p}_4) \delta^{(4)}(\vec{q} + \vec{w} - \vec{l} - \vec{u}), \end{aligned} \quad (73)$$

where n is the number of independent loop momenta and S is the symmetry factor.

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4 \quad (74)$$

$$\begin{aligned} \star &= -\frac{\lambda^4}{(2\pi)^{4n}} S \int \frac{d^4p d^4k d^4q d^4l d^4w d^4u}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{l}^2 - m^2 + i\epsilon} \\ &\quad \frac{\delta^{(4)}(\vec{q} + \vec{w} - \vec{l} - \vec{u}) \delta^{(4)}(\vec{k} - \vec{q} - \vec{K})}{\vec{k}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{p} + \vec{u} - \vec{k} - \vec{w}) \delta^{(4)}(\vec{P} + \vec{l} - \vec{p})}{\vec{w}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{P} + \vec{l} - \vec{p})}{\vec{u}^2 - m^2 + i\epsilon} \end{aligned} \quad (75)$$

I explain the rule for calculating symmetry factors in my Real Scalar Field Notes. The numerator combinatorics are tricky this time, so I show them in Fig. 7.

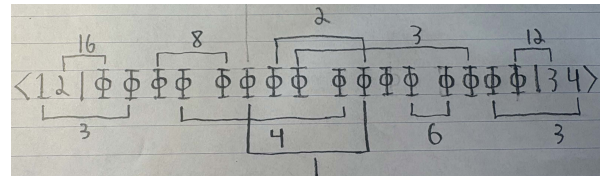


FIG. 7. Determining the symmetry factor numerator.

$$S = \frac{16 \times 12 \times 8 \times 6 \times 4 \times 3^3 \times 2}{(4!)^5} = \frac{1}{4} \quad (76)$$

$$\star = -\frac{\lambda^4}{4(2\pi)^{4n}} \int \frac{d^4p d^4k d^4q d^4l d^4w d^4u}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{l}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{q} + \vec{w} - \vec{l} - \vec{u})}{\vec{k}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{k} - \vec{q} - \vec{K})}{\vec{q}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{p} + \vec{u} - \vec{k} - \vec{w})}{\vec{w}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{P} + \vec{l} - \vec{p})}{\vec{u}^2 - m^2 + i\epsilon} \quad (77)$$

Four delta functions imply 3 independent loop momenta:
 $n = 3$.

$$\star = -\frac{\lambda^4}{4(2\pi)^{12}} \int \frac{d^4p d^4k d^4q d^4l d^4w d^4u}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{l}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{q} + \vec{w} - \vec{l} - \vec{u})}{\vec{k}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{k} - \vec{q} - \vec{K})}{\vec{q}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{p} + \vec{u} - \vec{k} - \vec{w})}{\vec{w}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{P} + \vec{l} - \vec{p})}{\vec{u}^2 - m^2 + i\epsilon} \quad (78)$$

$$\star = -\frac{\lambda^4}{4(2\pi)^{12}} \int \frac{d^4p d^4k d^4q d^4w d^4u}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{k}^2 - m^2 + i\epsilon} \frac{1}{(\vec{q} + \vec{w} - \vec{u})^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{k} - \vec{q} - \vec{K})}{\vec{q}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{p} + \vec{u} - \vec{k} - \vec{w})}{\vec{w}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{P} + \vec{q} + \vec{w} - \vec{u} - \vec{p})}{\vec{u}^2 - m^2 + i\epsilon} \quad (79)$$

$$\star = -\frac{\lambda^4}{4(2\pi)^{12}} \int \frac{d^4p d^4k d^4q d^4w}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{k}^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{P} + \vec{q} - \vec{k})}{\vec{w}^2 - m^2 + i\epsilon} \frac{1}{(\vec{p} - \vec{P})^2 - m^2 + i\epsilon} \frac{\delta^{(4)}(\vec{k} - \vec{q} - \vec{K})}{\vec{q}^2 - m^2 + i\epsilon} \frac{1}{(\vec{P} + \vec{q} + \vec{w} - \vec{p})^2 - m^2 + i\epsilon} \quad (80)$$

$$\star = -\frac{\lambda^4 \delta^{(4)}(\vec{P} - \vec{K})}{4(2\pi)^{12}} \int \frac{d^4p d^4k d^4w}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{k}^2 - m^2 + i\epsilon} \frac{1}{\vec{w}^2 - m^2 + i\epsilon} \frac{1}{(\vec{p} - \vec{P})^2 - m^2 + i\epsilon} \frac{1}{(\vec{k} - \vec{K})^2 - m^2 + i\epsilon} \frac{1}{(\vec{P} + \vec{k} - \vec{K} + \vec{w} - \vec{p})^2 - m^2 + i\epsilon} \quad (81)$$

$$\star = -\frac{\lambda^4 \delta^{(4)}(\vec{P} - \vec{K})}{4(2\pi)^{12}} \int \frac{d^4p d^4k d^4w}{\vec{p}^2 - m^2 + i\epsilon} \frac{1}{\vec{k}^2 - m^2 + i\epsilon} \frac{1}{\vec{w}^2 - m^2 + i\epsilon} \frac{1}{(\vec{p} - \vec{P})^2 - m^2 + i\epsilon} \frac{1}{(\vec{k} - \vec{P})^2 - m^2 + i\epsilon} \frac{1}{(\vec{k} + \vec{w} - \vec{p})^2 - m^2 + i\epsilon} \quad (82)$$

I use the convention in Schwartz, so my solution differs from the official solution by the factor $(2\pi)^4$. I use the shorthand

$$\mathcal{F}(p) = \frac{1}{p^2 - m^2 + i\epsilon}. \quad (83)$$

In the past I always put an arrow on 4-vectors, $\vec{p}$. But for this problem I switch to the shorthand $p = \vec{p}$.

$$\star = -\frac{\lambda^4 \delta^{(4)}(P - K)}{4(2\pi)^{12}} \int d^4p d^4k d^4w \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(w) \mathcal{F}(p - P) \mathcal{F}(k - P) \mathcal{F}(k + w - p) \quad (84)$$

This is correct but it does not match the official solution. I will play around with dummy variables until it does. Because I am integrating over all values of momentum and because each integration is independent I can swap $p \leftrightarrow -p$ as I please.

$$\star = -\frac{\lambda^4 \delta^{(4)}(P - K)}{4(2\pi)^{12}} \int d^4p d^4k d^4w \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(w) \mathcal{F}(p - P) \mathcal{F}(k - P) \mathcal{F}(k + w - p) \quad (85)$$

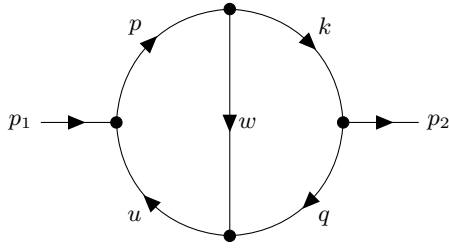
$$\star = -\frac{\lambda^4 \delta^{(4)}(P - K)}{4(2\pi)^{12}} \int d^4p d^4k d^4w \mathcal{F}(-p) \mathcal{F}(-k) \mathcal{F}(w) \mathcal{F}(-p - P) \mathcal{F}(-k - P) \mathcal{F}(w + p - k) \quad (86)$$

$$\mathcal{F}(p) = \mathcal{F}(-p) \quad (87)$$

$$\star = -\frac{\lambda^4 \delta^{(4)}(P - K)}{4(2\pi)^{12}} \int d^4p d^4k d^4w \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(w) \mathcal{F}(p + P) \mathcal{F}(k + P) \mathcal{F}(w + p - k) \quad (88)$$

2.3. b

I label this diagram $\mathfrak{D}$ and show it in Fig. 8. Remember that I am not putting the arrow over 4-vectors anymore.

FIG. 8. $\mathbb{D}$

$$\mathbb{D} = \frac{(-ig)^4 i^5}{(2\pi)^{4n}} S \int d^4 p d^4 k d^4 q d^4 u d^4 w \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(u) \mathcal{F}(w) \delta^{(4)}(p_1 + u - p) \delta^{(4)}(p - w - k) \delta^{(4)}(k - p_2 - q) \delta^{(4)}(q + w - u) \quad (89)$$

$$p_1 = P, \quad p_2 = K \quad (90)$$

$$\mathbb{D} = \frac{ig^4}{(2\pi)^{4n}} S \int d^4 p d^4 k d^4 q d^4 u d^4 w \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(u) \mathcal{F}(w) \delta^{(4)}(P + u - p) \delta^{(4)}(p - w - k) \delta^{(4)}(k - K - q) \delta^{(4)}(q + w - u) \quad (91)$$

I determine the symmetry factor S using the rule I described in my Real Scalar Field Notes, but I do not include the diagram showing the contractions this time.

$$S = \frac{12 \times 9 \times 6 \times 4 \times 3 \times 2}{(3!)^4 4!} = \frac{1}{2} \quad (92)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^{4n}} \int d^4 k d^4 q d^4 u d^4 w \mathcal{F}(P + u) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(u) \mathcal{F}(w) \delta^{(4)}(P + u - w - k) \delta^{(4)}(k - K - q) \delta^{(4)}(q + w - u) \quad (93)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^{4n}} \int d^4 k d^4 u d^4 w \mathcal{F}(P + u) \mathcal{F}(k) \mathcal{F}(k - K) \mathcal{F}(u) \mathcal{F}(w) \delta^{(4)}(P + u - w - k) \delta^{(4)}(k - K + w - u) \quad (94)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^{4n}} \delta^{(4)}(P - K) \int d^4 k d^4 u \mathcal{F}(P + u) \mathcal{F}(k) \mathcal{F}(k - K) \mathcal{F}(u) \mathcal{F}(P + u - k) \quad (95)$$

$$n = 2 \quad (96)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4 k d^4 p \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(P + p) \mathcal{F}(k - K) \mathcal{F}(P + p - k) \quad (97)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4 k d^4 p \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(P + p) \mathcal{F}(k - P) \mathcal{F}(P + p - k) \quad (98)$$

I use substitution to bring this into agreement with the official solution.

$$v = -p, \quad u = P + p - k \quad (99)$$

Forget that these are 4-vectors and transform coordinates like you would in 2D Euclidean space.

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4 v d^4 u \det \mathbf{J} \mathcal{F}(-v) \mathcal{F}(P - v - u) \mathcal{F}(P - v) \mathcal{F}(-v - u) \mathcal{F}(u) \quad (100)$$

$\mathbf{J}$ is the Jacobian. The integral before and after is over all of double momentum space: because the transformation is linear I don't need to worry about limits of integration.

$$\mathbf{J} = \begin{pmatrix} \frac{\partial k}{\partial u} & \frac{\partial k}{\partial v} \\ \frac{\partial p}{\partial u} & \frac{\partial p}{\partial v} \end{pmatrix} \quad (101)$$

Each element in this Jacobian is actually a 4×4 matrix.

$$p = -v, \quad k = P - v - u \quad (102)$$

$$\mathbf{J} = \begin{pmatrix} -I & -I \\ Z & -I \end{pmatrix} \quad (103)$$

I is the identity matrix and Z is the zero matrix. I get the determinant using Mathematica

$$\det \mathbf{J} = 1. \quad (104)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4 v d^4 u \mathcal{F}(v) \mathcal{F}(u) \mathcal{F}(v + u) \mathcal{F}(P - v) \mathcal{F}(P - v - u) \quad (105)$$

$$\mathbb{D} = \frac{ig^4}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4 p d^4 k \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(p + k) \mathcal{F}(P - p) \mathcal{F}(P - p - k) \quad (106)$$

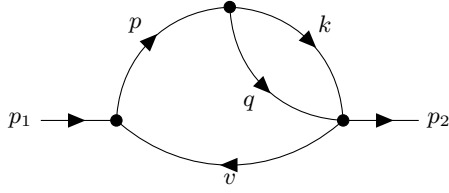


FIG. 9. ☿

2.4. c

I label the diagram ☿ and show it in Fig. 9.

$$P = p_1, \quad K = p_2 \quad (107)$$

$$\begin{aligned} \text{☿} &= \frac{(-ig)^2(-i\lambda)i^4}{(2\pi)^{4n}} S \int d^4p d^4k d^4q d^4v \mathcal{F}(p) \mathcal{F}(k) \\ &\mathcal{F}(q) \mathcal{F}(v) \delta^{(4)}(P + v - p) \delta^{(4)}(p - k - q) \delta^{(4)}(k + q - v - K) \end{aligned} \quad (108)$$

$$S = \frac{6^2 \times 4 \times 3 \times 2}{4!(3!)^2 2!} = \frac{1}{2} \quad (109)$$

$$\begin{aligned} \text{☿} &= -\frac{g^2 \lambda i}{2(2\pi)^{4n}} \int d^4p d^4k d^4q d^4v \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(v) \\ &\delta^{(4)}(P + v - p) \delta^{(4)}(p - k - q) \delta^{(4)}(k + q - v - K) \end{aligned} \quad (110)$$

$$\begin{aligned} \text{☿} &= -\frac{g^2 \lambda i}{2(2\pi)^{4n}} \int d^4p d^4k d^4q \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \\ &\mathcal{F}(k + q - K) \delta^{(4)}(P - K + k + q - p) \delta^{(4)}(p - k - q) \end{aligned} \quad (111)$$

$$\begin{aligned} \text{☿} &= -\frac{g^2 \lambda i}{2(2\pi)^{4n}} \delta^{(4)}(P - K) \int d^4p d^4k \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(p - k) \\ &\mathcal{F}(p - K) \end{aligned} \quad (112)$$

$$n = 2 \quad (113)$$

$$\begin{aligned} \text{☿} &= -\frac{g^2 \lambda i}{2(2\pi)^8} \delta^{(4)}(P - K) \int d^4p d^4k \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(p - k) \\ &\mathcal{F}(p - P) \end{aligned} \quad (114)$$

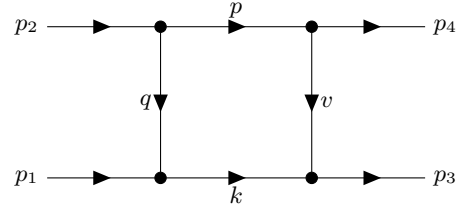


FIG. 10. ☿

2.5. d

I label the diagram ☿ and show it in Fig. 10.

$$P = p_1 + p_2, \quad K = p_3 + p_4 \quad (115)$$

$$\begin{aligned} \text{☿} &= \frac{(-ig)^4 i^4}{(2\pi)^{4n}} S \int d^4p d^4k d^4q d^4v \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(v) \\ &\delta^{(4)}(p_2 - p - q) \delta^{(4)}(p_1 + q - k) \delta^{(4)}(p - v - p_4) \delta^{(4)}(k + v - p_3) \end{aligned} \quad (116)$$

$$S = \frac{12 \times 9 \times 6 \times 4 \times 3 \times 2^2}{4!(3!)^4} = 1 \quad (117)$$

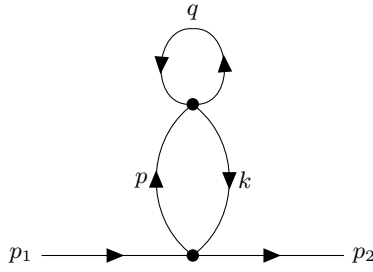
$$\begin{aligned} \text{☿} &= \frac{g^4}{(2\pi)^{4n}} \int d^4p d^4k d^4q d^4v \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(v) \\ &\delta^{(4)}(p_2 - p - q) \delta^{(4)}(p_1 + q - k) \delta^{(4)}(p - v - p_4) \delta^{(4)}(k + v - p_3) \end{aligned} \quad (118)$$

$$\begin{aligned} \text{☿} &= \frac{g^4}{(2\pi)^{4n}} \int d^4p d^4k d^4q \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \mathcal{F}(p - p_4) \\ &\delta^{(4)}(p_2 - p - q) \delta^{(4)}(p_1 + q - k) \delta^{(4)}(k + p - K) \end{aligned} \quad (119)$$

$$\begin{aligned} \text{☿} &= \frac{g^4}{(2\pi)^{4n}} \int d^4p d^4k \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(p_2 - p) \mathcal{F}(p - p_4) \\ &\delta^{(4)}(P - p - k) \delta^{(4)}(k + p - K) \end{aligned} \quad (120)$$

$$\begin{aligned} \text{☿} &= \frac{g^4}{(2\pi)^{4n}} \delta^{(4)}(P - K) \int d^4p d^4k \mathcal{F}(p) \mathcal{F}(P - p) \\ &\mathcal{F}(p_2 - p) \mathcal{F}(p - p_4) \end{aligned} \quad (121)$$

$$n = 1 \quad (122)$$

FIG. 11. $\mathfrak{h}$

$$\mathfrak{V} = \frac{g^4}{(2\pi)^4} \delta^{(4)}(P - K) \int d^4p d^4k \mathcal{F}(-p) \mathcal{F}(P + p) \mathcal{F}(p_2 + p) \mathcal{F}(-p - p_4) \quad (123)$$

$$\mathfrak{V} = \frac{g^4}{(2\pi)^4} \delta^{(4)}(P - K) \int d^4p d^4k \mathcal{F}(p) \mathcal{F}(P + p) \mathcal{F}(p_2 + p) \mathcal{F}(p + p_4) \quad (124)$$

2.6. e

I label the diagram $\mathfrak{h}$ and show it in Fig. 11.

$$P = p_1, \quad K = p_2 \quad (125)$$

$$\mathfrak{h} = \frac{(-i\lambda)^2 i^3}{(2\pi)^{4n}} S \int d^4p d^4k d^4q \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \delta^{(4)}(P + k - p - K) \delta^{(4)}(p + q - q - k) \quad (126)$$

$$S = \frac{8 \times 4 \times 3^2}{(4!)^2 2!} = \frac{1}{4} \quad (127)$$

$$\mathfrak{h} = \frac{i\lambda^2}{4(2\pi)^{4n}} \int d^4p d^4k d^4q \mathcal{F}(p) \mathcal{F}(k) \mathcal{F}(q) \delta^{(4)}(P + k - p - K) \delta^{(4)}(p - k) \quad (128)$$

$$\mathfrak{h} = \frac{i\lambda^2}{4(2\pi)^{4n}} \delta^{(4)}(P - K) \int d^4p d^4q \mathcal{F}(p)^2 \mathcal{F}(q) \quad (129)$$

$$n = 2 \quad (130)$$

$$\mathfrak{h} = \frac{i\lambda^2}{4(2\pi)^8} \delta^{(4)}(P - K) \int d^4p d^4q \mathcal{F}(p)^2 \mathcal{F}(q) \quad (131)$$

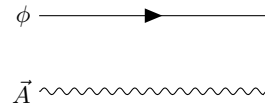


FIG. 12. Scalar QED Feynman lines.

3. 3

For this problem I use the notation and conventions of my QED Notes.

3.1. Scalar QED Feynman Rules

The Lagrangian is given in Peskin & Schroeder Problem 9.1

$$\mathcal{L} = -\frac{F^2}{4} + |D\phi|^2 - m^2|\phi|^2. \quad (132)$$

where $F^2 = F_{\mu\nu}F^{\mu\nu}$ and $|D\phi|^2 = (D_\mu\phi)^*(D^\mu\phi)$. Scalar QED couples the real electromagnetic field to a fictional complex scalar field (Fig. 12). The covariant derivative is $D_\mu = \partial_\mu + ieA_\mu$. I Expand the Lagrangian.

$$\mathcal{L} = -\frac{F^2}{4} - m^2|\phi|^2 + (\partial_\mu\phi + ieA_\mu\phi)^*(\partial^\mu\phi + ieA^\mu\phi) \quad (133)$$

$$\mathcal{L} = -\frac{F^2}{4} - m^2|\phi|^2 + (\partial_\mu\phi^* - ieA_\mu\phi^*)(\partial^\mu\phi + ieA^\mu\phi) \quad (134)$$

$$\mathcal{L} = (\partial_\mu\phi^*)(\partial^\mu\phi) - \frac{F^2}{4} - m^2|\phi|^2 - (ie)^2 A_\mu A^\mu |\phi|^2 + ie(A^\mu\phi\partial_\mu\phi^* - A_\mu\phi^*\partial^\mu\phi) \quad (135)$$

$$(\partial_\mu\phi^*)(\partial^\mu\phi) = |\partial\phi|^2, \quad \vec{A} = \vec{A}^2 \quad (136)$$

$$\mathcal{L} = |\partial\phi|^2 - \frac{F^2}{4} - m^2|\phi|^2 + e^2 \vec{A}^2 |\phi|^2 + ieA^\mu(\phi\partial_\mu\phi^* - \phi^*\partial_\mu\phi) \quad (137)$$

The free Lagrangian is

$$\mathcal{L}_0 = |\partial\phi|^2 - \frac{F^2}{4} - m^2|\phi|^2. \quad (138)$$

$$\mathcal{L} = \mathcal{L}_0 + e^2 \vec{A}^2 |\phi|^2 + ieA^\mu(\phi\partial_\mu\phi^* - \phi^*\partial_\mu\phi) \quad (139)$$

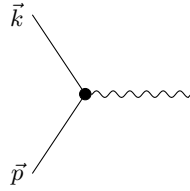


FIG. 13. The 3-way vertex contributes a factor of $ie(\pm\vec{p} \pm \vec{k})$, where $-$ is for particles and $+$ for antiparticles, see Schwartz Eqs. 9.17 - 9.20.

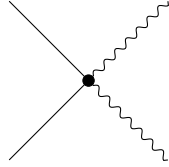


FIG. 14. The 4-way vertex contributes $2ie^2\eta_{\mu\nu}$ (Schwartz Eq. 9.23).

This matches the Lagrangian in Schwartz (Eq. 9.11). The propagators are

$$\boxed{\vec{A} \sim -\frac{i\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon}} \quad \boxed{\phi \sim \frac{i}{\vec{p}^2 - m^2 + i\epsilon}}. \quad (140)$$

where I put the photon propagator (Schwartz Eq. 9.13) in the Feynman gauge and $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. The scalar field operator is

$$\phi(\vec{x}) = \int d^3p \frac{a_p e^{-i\vec{p} \cdot \vec{x}} + b_p^\dagger e^{i\vec{p} \cdot \vec{x}}}{\sqrt{2E_p}(2\pi)^3}, \quad (141)$$

where $E_p = \sqrt{p^2 + m^2}$ and $p = |\mathbf{p}|$. I follow Schwartz and call what a annihilates particles and what b annihilates antiparticles. There are two interaction vertices: a 3-way (Fig. 13) and a 4-way (Fig. 14).

3.2. QED Feynman Rules

The free Lagrangian is

$$\boxed{\mathcal{L}_0 = \bar{\psi}(i\cancel{\partial} - m)\psi - \frac{F^2}{4}}. \quad (142)$$

The QED Lagrangian is

$$\boxed{\mathcal{L} = \mathcal{L}_0 - e\bar{\psi}\gamma^\mu\psi A_\mu}. \quad (143)$$

I represent the two fields using the lines in Fig. 15. The propagators are

$$\boxed{\vec{A} \sim -\frac{i\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon}} \quad \boxed{\psi \sim i\frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon}}. \quad (144)$$

I show the QED vertex in Fig. 16.

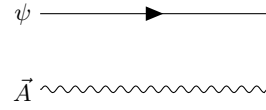


FIG. 15. QED Feynman lines.

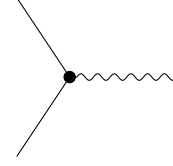


FIG. 16. The QED vertex contributes the factor $-ie\gamma^\mu$.

3.3. a

For the rest of this problem I stop putting arrows on 4-vectors. I draw scalar QED diagram $\star$ in Fig. 17. I determine the symmetry factor using the combinatorics in Fig. 18. The key is to assign each vertex a particle $(-)$ and antiparticle $(+)$. This captures the presence of both ϕ and ϕ^* in each vertex of Eq. 139. The diagram has two 4-way vertices, each with 2 identical A fields: that explains the denominator. Because particles and antiparticles are different swapping them does not contribute to the symmetry factor.

$$S = \frac{2 \times 2}{2!(2!)^2} = \frac{1}{2} \quad (145)$$

$$K = p_1 + p_2 \quad (146)$$

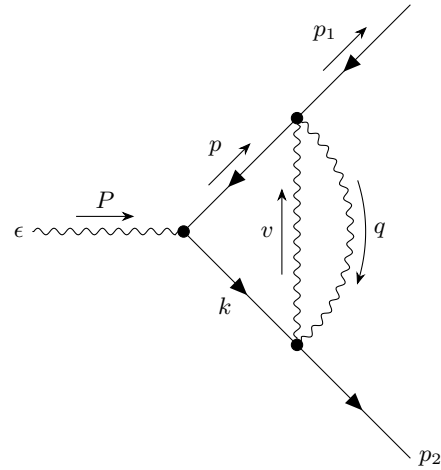
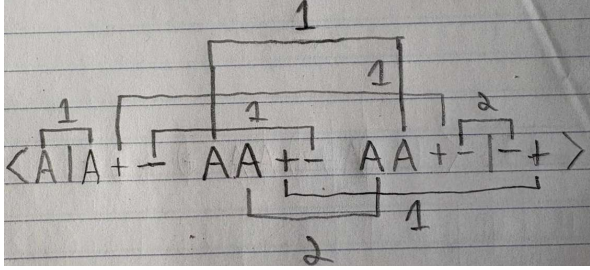


FIG. 17. Feynman diagram $\star$. I only include momentum arrows for antiparticles and photons. The scalar particles are charged and charge must be conserved at each vertex.

I am supposed to remove the external polarization $\epsilon(P)$ and leave an uncontracted index (μ) in its place.



$$\star = -\frac{2ie^5}{(2\pi)^{4n}} \eta^{\alpha\beta} \eta_{\alpha\gamma} \eta_{\beta\rho} \eta^{\gamma\rho} \int d^4p d^4k d^4v d^4q \frac{1}{p^2 - m^2 + i\epsilon} \frac{1}{k^2 - m^2 + i\epsilon} \frac{1}{v^2 + i\epsilon} \frac{p^\mu - k^\mu}{q^2 + i\epsilon} \delta^{(4)}(k + q - v - p_2) \delta^{(4)}(P - p - k) \delta^{(4)}(p + v - q - p_1) \quad (149)$$

$$\eta^{\alpha\beta} \eta_{\alpha\gamma} \eta^{\gamma\rho} \eta_{\beta\rho} = \delta_\gamma^\beta \delta_\beta^\gamma = 4 \quad (150)$$

$$\mathcal{F}(p) = \frac{1}{p^2 - m^2 + i\epsilon}, \quad \mathcal{G}(p) = \frac{1}{p^2 + i\epsilon} \quad (151)$$

$$\star = -\frac{8ie^5}{(2\pi)^{4n}} \int d^4p d^4k d^4v d^4q (p^\mu - k^\mu) \mathcal{F}(p) \mathcal{F}(k) \mathcal{G}(v) \mathcal{G}(q) \delta^{(4)}(k + q - v - p_2) \delta^{(4)}(P - p - k) \delta^{(4)}(p + v - q - p_1) \quad (152)$$

FIG. 18. Symmetry factor for Fig. 17.

$$\star = \frac{S}{(2\pi)^{4n}} \int d^4p d^4k d^4v d^4q \epsilon^\mu(P) i e (p_\mu - k_\mu) \frac{i}{p^2 - m^2 + i\epsilon} \frac{i}{k^2 - m^2 + i\epsilon} 2ie^2 \eta^{\alpha\beta} \frac{-i\eta_{\alpha\gamma}}{v^2 + i\epsilon} \frac{-i\eta_{\beta\rho}}{q^2 + i\epsilon} 2ie^2 \eta^{\gamma\rho} \delta^{(4)}(P - p - k) \delta^{(4)}(p + v - q - p_1) \delta^{(4)}(k + q - v - p_2) \quad (147)$$

$$\star = -\frac{8ie^5}{(2\pi)^{4n}} \int d^4p d^4v d^4q (2p^\mu - P^\mu) \mathcal{F}(p) \mathcal{F}(P - p) \mathcal{G}(v) \mathcal{G}(q) \delta^{(4)}(P - p + q - v - p_2) \delta^{(4)}(p + v - q - p_1) \quad (153)$$

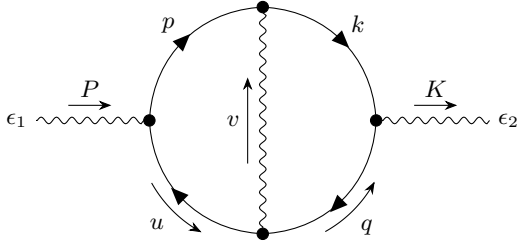
$$\star = -\frac{8ie^5}{(2\pi)^{4n}} \delta^{(4)}(P - p_1 - p_2) \int d^4p d^4q (2p^\mu - P^\mu) \mathcal{F}(p) \mathcal{G}(q) \mathcal{F}(P - p) \mathcal{G}(P - p_2 - p + q) \quad (154)$$

$$\star = -\frac{8ie^5}{(2\pi)^{4n}} \delta^{(4)}(P - p_1 - p_2) \int d^4p d^4q (2p^\mu - P^\mu) \mathcal{F}(p) \mathcal{G}(q) \mathcal{F}(P - p) \mathcal{G}(P - p_2 - p + q) \quad (155)$$

$$\star = -\frac{2ie^5}{(2\pi)^{4n}} \eta^{\alpha\beta} \eta_{\alpha\gamma} \eta_{\beta\rho} \eta^{\gamma\rho} \epsilon_\mu(P) \int d^4p d^4k d^4v d^4q \frac{1}{p^2 - m^2 + i\epsilon} \frac{1}{k^2 - m^2 + i\epsilon} \frac{1}{v^2 + i\epsilon} \frac{p^\mu - k^\mu}{q^2 + i\epsilon} \delta^{(4)}(k + q - v - p_2) \delta^{(4)}(P - p - k) \delta^{(4)}(p + v - q - p_1) \quad (148)$$

$$P = K, \quad P - p_2 = p_1, \quad n = 2 \quad (156)$$

$$\star = \frac{8e^5}{i(2\pi)^8} \delta^{(4)}(P - K) \int d^4p d^4q (2p^\mu - P^\mu) \mathcal{F}(p) \mathcal{G}(q) \mathcal{F}(P - p) \mathcal{G}(p_1 - p + q) \quad (157)$$

FIG. 19. $\odot$

$$\star = \frac{\delta^{(4)}(P-K)}{32i\pi^8} e^5 \int d^4p d^4q (2p^\mu - P^\mu) \mathcal{F}(p) \mathcal{G}(q) \mathcal{F}(P-p) \mathcal{G}(p_1 - p + q) \quad (158)$$

$$\star = \frac{\delta^{(4)}(P-K)}{32\pi^8} ie^5 \int d^4p d^4q (2p^\mu + P^\mu) \mathcal{F}(-p) \mathcal{G}(q) \mathcal{F}(P+p) \mathcal{G}(p_1 + p + q) \quad (159)$$

$$\mathcal{F}(p) = \mathcal{F}(-p) \quad (160)$$

$$\star = \frac{\delta^{(4)}(P-K)}{32\pi^8} ie^5 \int d^4p d^4q (2p^\mu + P^\mu) \mathcal{F}(p) \mathcal{G}(q) \mathcal{F}(P+p) \mathcal{G}(p_1 + p + q) \quad (161)$$

3.4. b

I call this QED diagram $\odot$ and draw it in Fig. 19. I show the combinatorics for determining the symmetry factor in Fig. 20.

$$S = \frac{4 \times 3 \times 2}{4!} = 1 \quad (162)$$

$$\begin{aligned} \odot &= -\frac{S}{(2\pi)^{4n}} \epsilon_{1\nu}(P) \epsilon_{2\mu}^*(K) \int d^4p d^4k d^4q d^4u d^4v \\ &\quad \text{Tr} \left[i \frac{\not{p} + m}{u^2 - m^2 + i\epsilon} (-ie\gamma^\beta) i \frac{\not{q} + m}{q^2 - m^2 + i\epsilon} (-ie\gamma^\mu) \right. \\ &\quad \left. i \frac{\not{k} + m}{k^2 - m^2 + i\epsilon} (-ie\gamma^\alpha) i \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} (-ie\gamma^\nu) \right] \frac{-i\eta_{\alpha\beta}}{v^2 + i\epsilon} \\ &\quad \delta^{(4)}(P-p-u) \delta^{(4)}(u-v-q) \delta^{(4)}(q+k-K) \delta^{(4)}(v+p-k) \end{aligned} \quad (163)$$

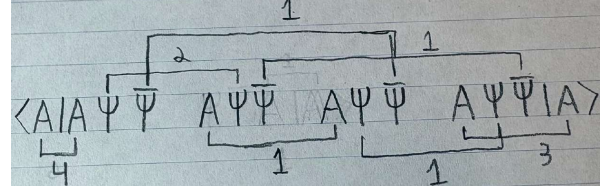


FIG. 20. Symmetry factor for Fig. 19.

Once again I drop the external polarizations.

$$\begin{aligned} \odot &= \frac{i^9 e^4}{(2\pi)^{4n}} \eta_{\alpha\beta} \int d^4p d^4k d^4q d^4u d^4v \mathcal{G}(v) \\ &\quad \text{Tr} [\mathcal{B}(u) \gamma^\beta \mathcal{B}(q) \gamma^\mu \mathcal{B}(k) \gamma^\alpha \mathcal{B}(p) \gamma^\nu] \\ &\quad \delta^{(4)}(P-p-u) \delta^{(4)}(u-v-q) \delta^{(4)}(q+k-K) \delta^{(4)}(v+p-k) \end{aligned} \quad (164)$$

$$\boxed{\mathcal{B}(p) = \frac{\not{p} + m}{p^2 - m^2 + i\epsilon}} \quad (165)$$

$$\begin{aligned} \odot &= \frac{ie^4}{(2\pi)^{4n}} \eta_{\alpha\beta} \int d^4p d^4k d^4q d^4u d^4v \mathcal{G}(v) \\ &\quad \text{Tr} [\mathcal{B}(u) \gamma^\beta \mathcal{B}(q) \gamma^\mu \mathcal{B}(k) \gamma^\alpha \mathcal{B}(p) \gamma^\nu] \\ &\quad \delta^{(4)}(P-p-u) \delta^{(4)}(u-v-q) \delta^{(4)}(q+k-K) \delta^{(4)}(v+p-k) \end{aligned} \quad (166)$$

$$\mathfrak{V} = \frac{ie^4}{(2\pi)^{4n}} \delta^{(4)}(P-K) \eta_{\alpha\beta} \eta_{\mu\nu} \int d^4p d^4q \mathcal{G}(p) \mathcal{G}(q) \gamma^\mu \mathcal{B}(P-p) \gamma^\alpha \mathcal{B}(P-p-q) \gamma^\beta \mathcal{B}(P-p) \gamma^\nu \quad (177)$$

$$n = 2 \quad (178)$$

$$\mathfrak{V} = \frac{ie^4}{(2\pi)^8} \delta^{(4)}(P-K) \eta_{\alpha\beta} \eta_{\mu\nu} \int d^4p d^4q \mathcal{G}(p) \mathcal{G}(q) \gamma^\mu \mathcal{B}(P-p) \gamma^\alpha \mathcal{B}(P-p-q) \gamma^\beta \mathcal{B}(P-p) \gamma^\nu \quad (179)$$

$$\mathfrak{V} = \frac{ie^4}{(2\pi)^8} \delta^{(4)}(P-K) \int d^4p d^4q \mathcal{G}(p) \mathcal{G}(q) \gamma^\mu \mathcal{B}(P-p) \eta_{\alpha\beta} \gamma^\alpha \mathcal{B}(P-p-q) \gamma^\beta \mathcal{B}(P-p) \eta_{\mu\nu} \gamma^\nu \quad (180)$$

$$\mathfrak{V} = \frac{ie^4}{(2\pi)^8} \delta^{(4)}(P-K) \int d^4p d^4q \mathcal{G}(p) \mathcal{G}(q) \gamma^\mu \mathcal{B}(P-p) \gamma_\beta \mathcal{B}(P-p-q) \gamma^\beta \mathcal{B}(P-p) \gamma_\mu \quad (181)$$

$$\mathfrak{V} = \frac{ie^4}{(2\pi)^8} \delta^{(4)}(P-K) \int d^4p d^4q \mathcal{G}(p) \mathcal{G}(q) \gamma^\mu \mathcal{B}(P-p) \gamma^\beta \mathcal{B}(P-p-q) \gamma_\beta \mathcal{B}(P-p) \gamma_\mu \quad (182)$$